

Sehat AI: A Healthcare Assitant for Every Pakistani

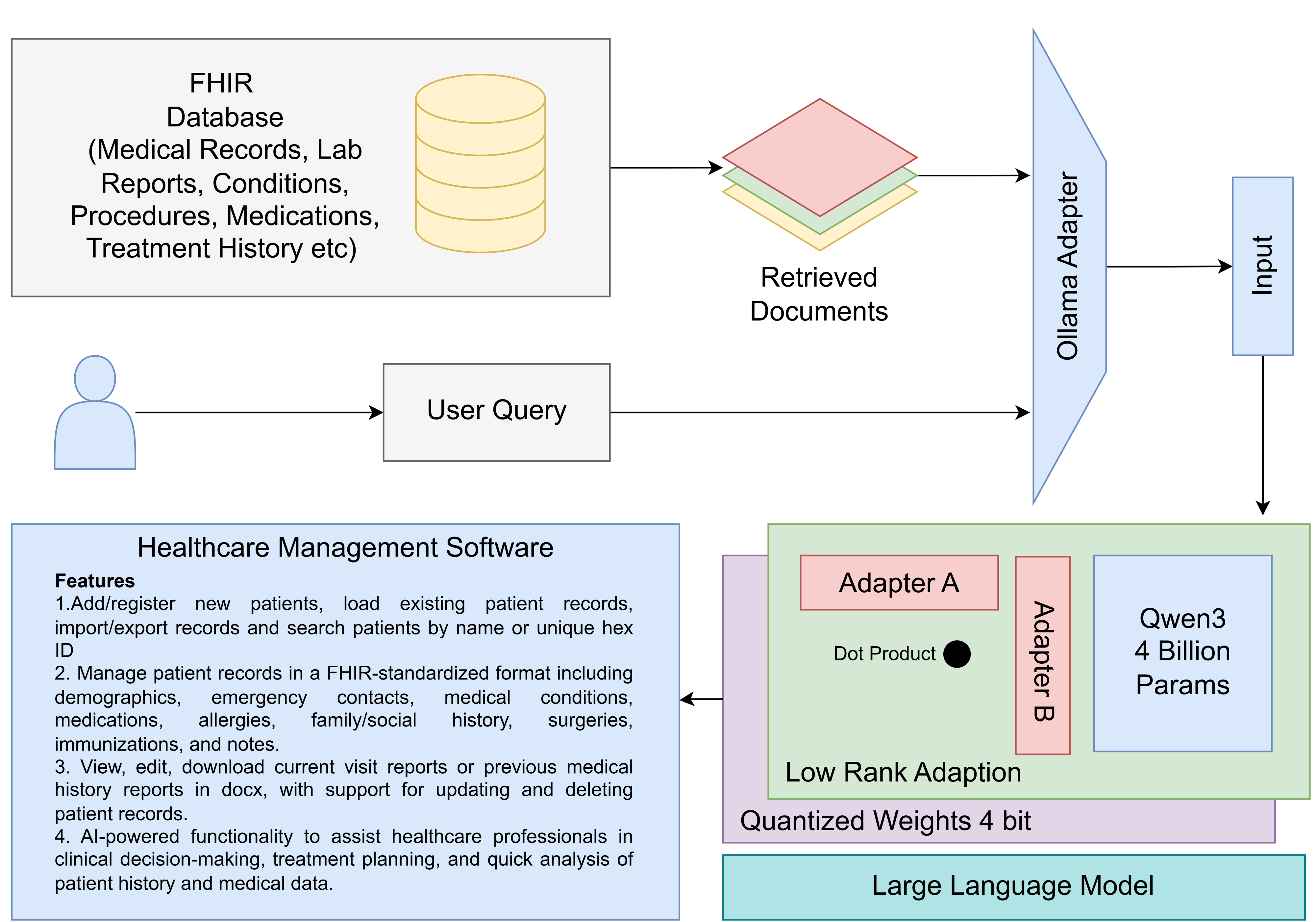
Abstract:

Pakistan’s healthcare system faces major challenges due to a shortage and uneven distribution of doctors, especially in rural and underserved regions. As a result, many people rely on unqualified practitioners, increasing the risk of unsafe and ineffective treatments. Artificial Intelligence and Large Language Models (LLMs) are transforming healthcare worldwide through AI-assisted diagnosis, treatment recommendations, hospital management, and remote patient care. Many developed countries have already integrated AI into clinical workflows to improve healthcare accessibility and efficiency. To address this gap, we propose Sehat AI, an AI-enabled healthcare management system designed for low-resource and low-bandwidth clinical environments. The system assists healthcare professionals with AI-powered clinical decision support, diagnosis, patient record management, medical history tracking, and report generation, helping improve efficiency, reduce workload, and provide better healthcare delivery.

Objectives:

- 1.Develop Sehat AI, an AI-enabled healthcare management system designed for clinical environments.
- 2.Provide healthcare professionals with AI-assisted clinical decision support including symptom analysis, diagnosis assistance, medicine recommendations, treatment planning, and patient history evaluation.
- 3.Implement a comprehensive EMR system supporting patient registration, visit tracking, medical history management, report generation, and simplified FHIR-standardized healthcare records.
- 4.Enable secure, low-bandwidth, and cost-effective healthcare infrastructure with offline AI integration using Ollama, ensuring no cloud dependency.

Block Diagram:



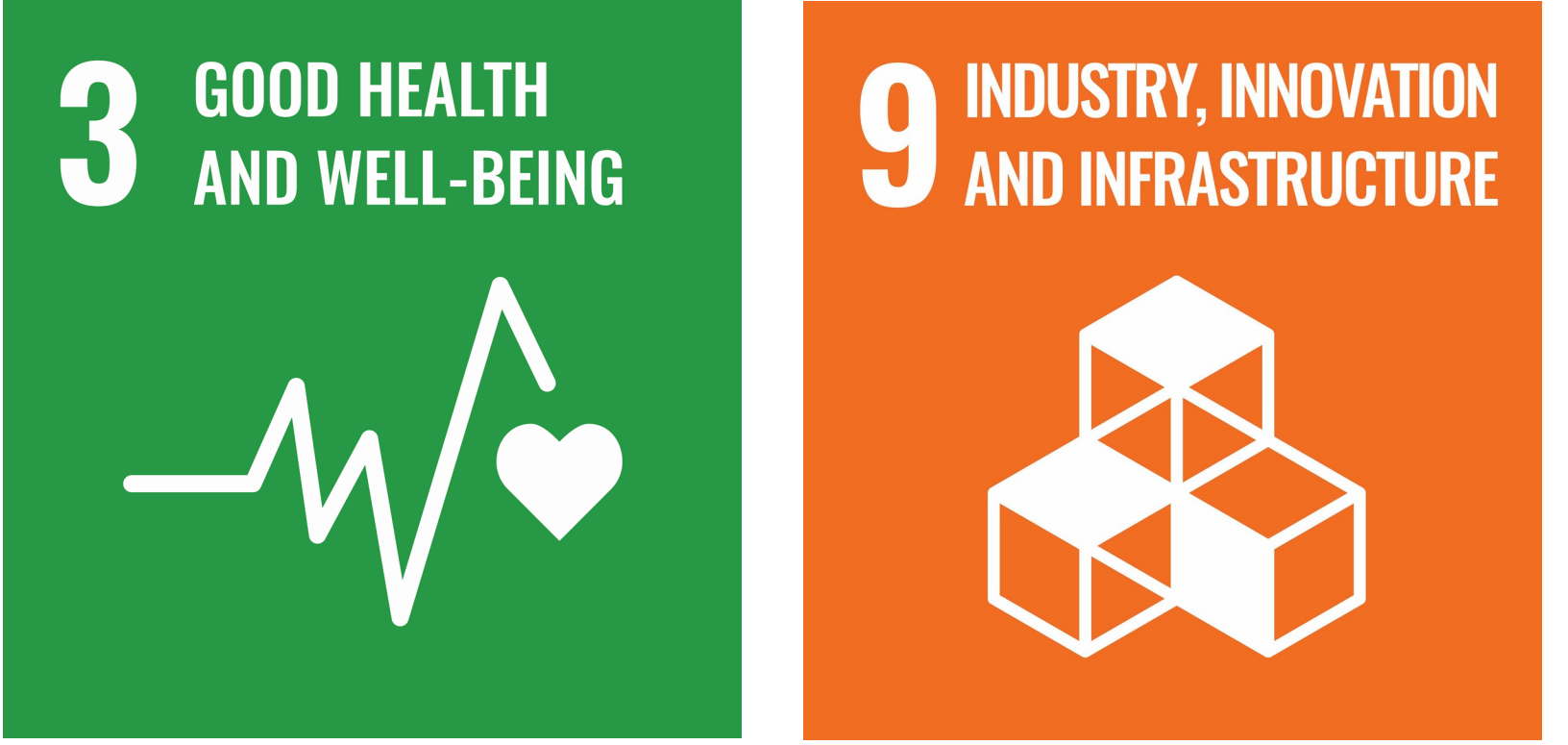
Results:

~8B Large Language Models									
Model	Size GB	MedQA	MedMCQA	PubMedQA	MMLU-Pro Health	MMLU-Pro Biology	GPQA Genetics	GPQA Molecular Biology	Average
BioMistral-7B	14.5	45	40.2	66.9	27.4	49.2	28.6	38.5	42.3
OpenBioLLM-8B	16.1	57.7	54.1	74.1	38.4	52.4	43.7	39.6	51.4
UltraMedical-8B	16.1	71.1	58.3	77.4	55.1	66.7	41.2	48.4	59.7
Mistral-7B-Instruct	14.5	48.2	44.6	59.5	33.7	53.6	30	46.1	45.1
Yi-1.5-9B-Chat	17.7	50.8	48.7	69.8	43.4	65.6	42.5	48.1	52.7
LLaMA-3.1-8B-Instruct	16.1	58.7	56	75.2	52.7	64.6	33.8	46.8	55.4
GLM-4-9B-Chat	19	58.9	49.8	73.5	45.5	65.4	53.8	41.6	55.5
Qwen2.5-7B-Instruct	15.2	57	55.6	72.7	50.6	70.2	36.2	49.7	56
Gemma2-9B	36.95	61.8	55.9	63.3	55.1	74.9	35	57.4	57.6
HuatuoGPT-o1-8B	16.1	72.6	60.4	79.2	58.7	68.2	48.8	59.7	63.9
w/o Stage2 (RL)	16.1	69	57.9	77.7	53.5	66.1	41.2	53.5	59.8
~4B MedQwen3 4 Bit Quantized (85.6% smaller almost 7 times less size)									
MedQwen3-4B-Q4	2.6	60.6	55.4	57	32	39.5	43.8	51.6	48.6

Applications:

- AI-assisted clinical decision support for diagnosis, treatment planning, and patient management.
- Secure electronic medical record management with patient history, reports, and FHIR-standardized data.
- Offline and low-cost healthcare management system for clinics, hospitals, and underserved areas.

SDGs:



Conclusion:

Sehat AI provides an AI-enabled healthcare management system designed for low-resource clinical environments. By combining offline AI assistance with electronic medical records, the system helps healthcare professionals improve patient management, clinical decision-making, and healthcare efficiency while ensuring privacy and accessibility without dependence on any cloud services.

The system offers a cost-effective solution for healthcare infrastructure, particularly in underserved regions. Sehat AI demonstrates the potential of integrating into Pakistan’s healthcare sector to support smarter, faster, and more efficient clinical workflows.



Student Names: Muhammad Abdul Rehman, Abdul Qadir Abbasi, Muhammad Tahawar Ali
Student Roll No.: 22-EE-40, 22-EE-44, 22-EE-72
Supervisor’s Name: Dr. Furqan Shoukat

DEPARTMENT OF ELECTRICAL ENGINEERING
UNIVERSITY OF ENGINEERING AND TECHNOLOGY TAXILA